

Paying When It Cannot Help: LLM Agents in a Collective-Risk Dilemma

Anonymous Author(s)

Abstract

Language-model agents increasingly act on behalf of people in settings where multiple agents share a common outcome. In many such settings, avoiding a collective loss requires each agent to give up some of its own resources. We ask whether such agents adjust their cooperation to the stakes and their partners, or instead follow a largely fixed default behaviour. Five production models play a collective-risk social dilemma in which six players pay into a shared pool over ten rounds; if the pool misses its target, a catastrophe destroys all remaining savings with a given probability. Four models pay about an equal share at every positive risk level, and four of the five also pay at zero risk, where paying cannot help. Explicitly stating that only the agent’s own cash matters reduces payment at zero risk for most models; simply making the prompt more neutral does not. Several models correctly identify which choice gives them the higher expected payoff at each risk level, yet barely change what they actually pay. Against fixed partners, no model consistently chooses the payoff-maximising best response, and in groups mixing different models, one agent that underpays in the final round can make an otherwise successful group fail. In payoff-based evolutionary selection, population size changes which model is favoured: with only one six-agent group, selection favours the model that pays least and reduces group success, whereas in larger randomly mixed populations it favours an exact fair-share model at moderate and high risk.

Keywords

LLM agents, collective-risk social dilemma, cooperation, best response, empirical game-theoretic analysis, multiagent evaluation

ACM Reference Format:

Anonymous Author(s). 2027. Paying When It Cannot Help: LLM Agents in a Collective-Risk Dilemma. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 10 pages.

1 Introduction

Language-model agents increasingly act on behalf of people in settings where several agents share a common outcome, such as spending budgets, drawing on shared compute or negotiating over common resources [7, 19]. In many such settings, avoiding a collective loss requires each agent to give up some of its own resources. We ask whether such agents adjust their cooperation to the situation or follow a largely fixed default. For a designer of such agents, this question splits into three. Does an agent’s cooperation depend

on what is at stake? Does it depend on what the other agents do? And which agents prevail when a system keeps or copies those that earn the most?

Success in self-play, common in multiagent benchmarks [22, 37], cannot answer these questions on its own. When every seat holds a copy of the same model, an agent that weighs the stakes and an agent that repeats a fixed rule can leave the same record of play. A success rate also hides how far a group is from the best it could have done: a group can reach its goal and still waste most of its money. And a prompt that states a group target and a number of rounds already suggests an equal split, so an agent that repeats that split looks cooperative on the usual summary measures.

We study these questions in the collective-risk social dilemma, a game introduced to study climate cooperation [31]. Players pay into a shared pool over several rounds, and if the pool misses a target, a catastrophe destroys everyone’s savings with probability p (Figure 1a). For risk-neutral players, group welfare is highest when nobody pays below $p = \frac{1}{2}$ and when the group reaches the target exactly above it, for example by each paying the *fair share*, the target split equally. The game also contains two situations in which paying cannot raise a player’s own cash: when there is no risk, and once the outcome is settled. Neither risk attitude nor concern for the group can explain payment in these situations, so they separate a default from a response to the stakes.

We introduce a game-theoretic evaluation that separates cooperative outcomes from incentive-sensitive behaviour by combining risk variation, dominated-action tests, exact best responses to fixed partners and selection in mixed populations, where each model is a strategy of an empirical game [48, 50]. We apply it to five production models, Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B and Grok 4.20, in four designs (Figure 1b), each compared with a benchmark computed from the game. We use standard equilibrium and finite-population tools, not a new solution concept. We call near-flat payment across risk levels under a fixed prompt a *default*; the term describes behaviour, not internal reasoning.

We report three sets of findings, one per question. (1) *Stakes*. Most models pay about the fair share at every positive risk level and also when there is no risk, and removing normative words changes neither result; stating that only the agent’s own cash matters cuts most zero-risk payment, but only GPT-5.6 Luna then moves from paying little at low risk to the fair share at high risk. Asked in separate calls, several models rank the two choices correctly and change that ranking with the risk, but they keep paying the same amount (Section 4.1). (2) *Partners*. Beside scripted partners who make every payment a loss, no model plays the best response of paying nothing, and two keep paying after the outcome is settled. In mixed tables, one partner that stops paying in the last round often breaks a table of exact fair-share players, while a table that overpays absorbs the shortfall (Section 4.2). (3) *Selection*. Population size changes who is selected under payoff-based imitation: a population



This work is licensed under a Creative Commons Attribution International 4.0 License.

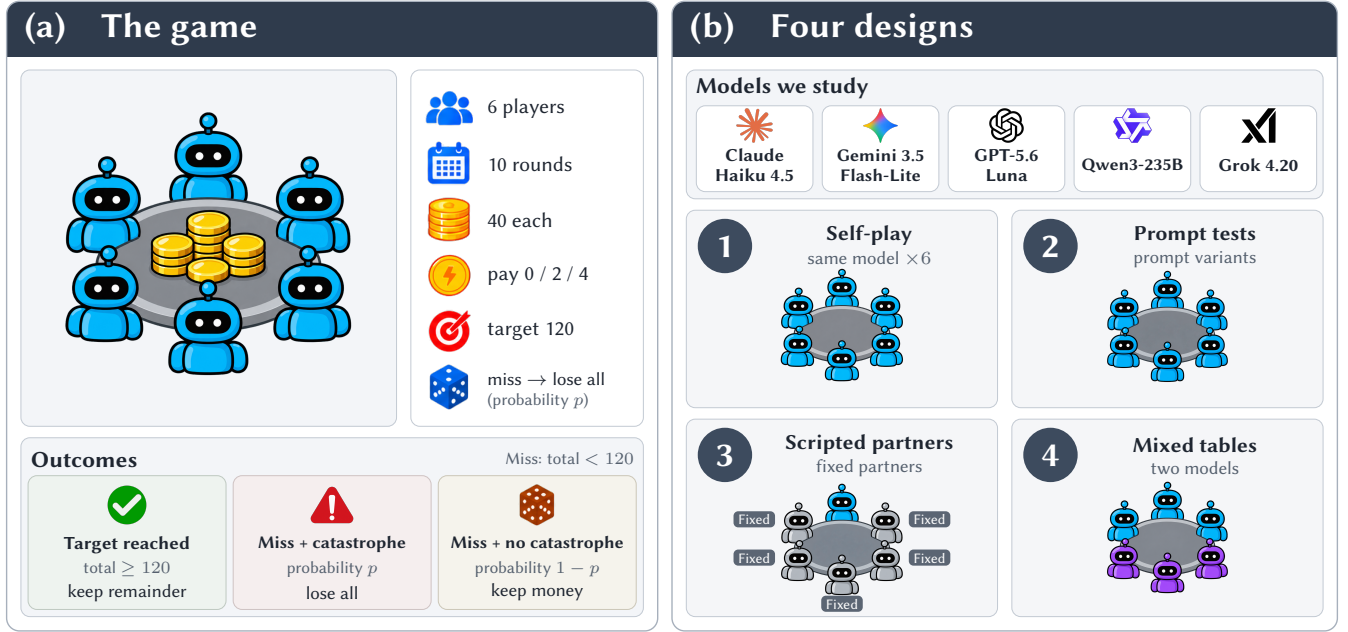


Figure 1: Overview. (a) The game (Section 3.1): six players hold 40 units each and pay 0, 2 or 4 into a shared pool in each of ten rounds; if the pool misses the target of 120, a catastrophe destroys all savings with probability p . (b) The five models and the four designs (Table 2): (1) self-play over the risk grid; (2) self-play under prompt and decoding controls, and with in-game value questions; (3) one model beside five scripted partners that follow fixed rules; (4) two models at one table, one of them in one to five seats. Selection (Section 3.3) reuses the base-prompt payoffs of designs 1, 2 and 4 and plays no new games.

of one table favours the smallest payer and lowers group success, whereas a larger randomly mixed population favours the exact fair-share player at moderate and high risk (Section 4.3).

For the five models we test, payment is a default rather than a response to the stakes. A default can work well beside copies of itself; whether it fails elsewhere depends on the partners, on whether the agent’s goal is stated, and on how agents are selected.

2 Related work

Collective risk and selection. The collective-risk dilemma was introduced to study why groups fail to avoid a shared disaster, and human groups reach the target mostly when the risk is high [31]; inequality and communication change how often they succeed [43]. Threshold public goods games have many equilibria [34], contributions decline over the rounds of repeated public goods games [3], and the risk of collective failure can turn a defection trap into a coordination problem [38]. In evolutionary simulations of the same game, constant fair sharers lose out to players who hold back early and compensate later [1]. Selection on payoff relative to a small group favours whoever does better than its neighbours [39], while selection between groups can favour cooperation [46]. Finite-population analyses price the incentives that sustain cooperation in this game [11] and show that maximising cooperation need not maximise welfare [21]. This literature predicts that behaviour moves with the risk, so a flat response is informative, and we reuse its selection theory to read our mixed tables.

Language models as strategic players. Language models have been studied as economic subjects [8, 15, 30] and in repeated two-player games [2, 16, 33]. Models tend to be more cooperative than people [30], and an even split can reflect a focal point rather than fairness [10]. The weight of framing relative to game structure differs by model [29], and models can hold a correct belief or plan and act against it [12, 40]. FAIRGAME runs language-model agents through standard games and finds outcomes that depend on model, language and persona [5]; later work extends it to a payoff-scaled prisoner’s dilemma and a multi-player public goods game [24].

Language models in social dilemmas. Societies of agents over-use a common resource [37], contributions to a public good can stay the same even when it returns nothing [22], and reasoning models free-ride [18]. Agents stand in for people in a collective-risk game [41] and for developers, regulators and users in an evolutionary model of AI regulation [4]. A study of 35 open models includes a collective-risk game at one risk level and finds play far from the group optimum [26]. In a binary collective-risk game with strategies written by three models, payoff-biased imitation drives larger groups to selfish, low-welfare outcomes [51], and prompts that evolve under individual selection collapse into defection while group selection keeps them prosocial [6]. A recent study audits rule recall, state tracking and payoff calculation before interpreting play in multi-player races to develop AI [36]. It compares play with an evolutionary benchmark and human data, and finds that correct recall can coexist with weak expected-payoff calculation. We differ in three ways: we vary the risk from zero to certainty and add two

Table 1: Symbols and terms.

Term	Meaning
Seat; table	One player position; the six seats of one game
Fair share	The target split equally: 20 per seat, or 2 per round
Exact split	Paying 2 in every one of the ten rounds
Pivot $p^* = \frac{1}{2}$	Risk above which reaching the target is optimal
$\text{opt}(p)$	Welfare-optimal expected payoff per seat
Dominated	Lowers the seat’s own payoff whatever others do
Settled	The history proves that more payment cannot change the outcome: the pool has reached T , or cannot reach it even if every seat pays 4 from now on

tests in which paying cannot raise the payer’s cash; we compute the exact best response to fixed partners; and we select among deployed models with payoffs from their real mixed tables, finding that at moderate and high risk the favoured model changes once the population exceeds one table.

Multiagent evaluation. Multiagent reinforcement learning studies cooperation in sequential social dilemmas [23, 28], and ad hoc teamwork and cooperative AI ask for agents that work with partners unlike themselves [9, 27, 42]. Empirical game-theoretic analysis treats trained agents as strategies of a meta-game [48, 50]. α -Rank ranks them by evolutionary dynamics [32], and evolutionary game theory is used to study emergent behaviour in multiagent systems [20]. Language models show distinct strategic fingerprints in tournaments with classic strategies [35], their cooperation evolves differently across base models [49], and cooperation sustained by repetition weakens when co-players differ [45]. In the collective-risk game, people who delegate to artificial agents succeed more often [14], and in theory adaptive players compensate for the effort of artificial agents [44]. We add a many-player dilemma with exact best responses to fixed partners, and measure this compensation in real models.

3 Game and evaluation design

3.1 Collective-risk game and benchmarks

Six players each start with $e = 40$ units, and the game lasts $R = 10$ rounds. In each round every player privately chooses a contribution $a_i^t \in \{0, 2, 4\}$ to a shared *pool*, and after each round all players see what every seat has paid. Let $x_i = \sum_t a_i^t$ be a player’s total and $X = \sum_i x_i$ the pool. The target is $T = 120$. A player keeps what it did not pay, but if the pool misses the target, a catastrophe destroys all savings with probability p (Table 1 lists the terms we use):

$$u_i = \begin{cases} e - x_i & \text{if } X \geq T, \\ 0 \text{ with probability } p, \text{ else } e - x_i & \text{if } X < T. \end{cases} \quad (1)$$

Proposition 1. *Consider the expected-payoff game in which each player picks a total $x_i \in \{0, 2, \dots, 40\}$, and let $p < 1$. A profile is a Nash equilibrium if and only if $x = 0$, or $X = T$ and $x_i \leq pe$ for every i . A target-reaching equilibrium exists if and only if $p \geq p^* = T/(6e) = \frac{1}{2}$. The welfare-optimal outcome is $x = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, and both are optimal at $p = p^*$, so the welfare-optimal mean payoff per player is $\text{opt}(p) = \max\{(1-p)e, e - T/6\}$; this welfare statement also holds at $p = 1$.*

Proof sketch. If $X < T$, any player with $x_i > 0$ gains by paying nothing, so the only equilibrium that misses the target is $x = 0$. If $X > T$, then $X \geq T + 2$ because totals are even, so a player can pay 2 less and still reach the target. If $X = T$, the best deviation is to pay nothing, worth $(1-p)e$ against $e - x_i$, which gives $x_i \leq pe$; summing over players gives $p \geq T/(6e)$, and conversely $x_i = 20$ for every i is an equilibrium when $p \geq \frac{1}{2}$. Welfare is $6e - X$ when the target is met and $(1-p)(6e - X)$ otherwise, so it peaks at $X = T$ or $X = 0$. The ten-round game, in which strategies can react to the history, has the same totals in pure-strategy Nash equilibrium; the supplementary material gives the full proof. We claim nothing about subgame perfection, and $\text{opt}(p)$ is a welfare bound that does not depend on the equilibrium concept. $\square$

What we compare play with. Proposition 1 gives a *symmetric welfare benchmark*: keep everything below the pivot and pay the fair share above it. Asymmetric allocations reaching exactly T attain the same group welfare. This benchmark is not a unique equilibrium or a prediction of individual play: paying nothing is also an equilibrium at every risk level. Paying is dominated in own cash at $p = 0$ and, for $p < 1$, after the outcome is settled. These tests do not require risk neutrality or beliefs about the partners. They do assume that more own cash is preferred; non-cash goals are not ruled out.

Proposition 2. *For two seats i and j in the same game, the difference in their expected payoffs over the lottery is $c(x_j - x_i)$, with $c = 1$ if the target was reached and $c = 1 - p$ otherwise.*

Both seats face the same pool and the same lottery, so only their own payments differ. Hence, for $p < 1$, whoever pays less at a table earns more, whatever the group achieves.

3.2 Models, protocol and measures

We test the five models of Section 1: one production model per provider, the kind plausible for deploying many agents at scale. Qwen3-235B is the instruction-tuned build without a thinking mode, and Grok 4.20 is its non-reasoning variant.

Prompts. In each round a seat receives one prompt with the rules, the full history of every seat’s payments, and a request for one contribution; the supplementary material gives every prompt word for word. The base prompt calls the pool a “climate account”, says that the group “must” reach the target, calls the loss a “disaster” and names the game as a collective-risk social dilemma. It never states the player’s goal, and it hints at the equal split twice. The prompt controls of Section 3.3 change exactly these parts.

Protocol. The running pool is not printed, so a seat that wants it must add up the history. The prompt is in English and no seat is given a persona. We read the move from the reply’s final CONTRIBUTION: line.

Measures. We report *expected* payoffs, the average a seat would earn if the catastrophe lottery were run many times on the same payments, and the share of $\text{opt}(p)$ that is lost. Proposition 1, $\text{opt}(p)$ and the selection fitness below all use expected payoffs, which remove the lottery noise without bias; agents never see the draw, and the supplementary material lists the realised draws. The unit of analysis is the game, since seats in one game share a history and a lottery. Intervals are 95% percentile bootstraps over games, tests

Table 2: Designs. A cell is one condition at one risk level: one model in self-play, one model with one kind of scripted partner, or one pair of models with one split of seats. Each cell holds ten games.

Design	Risk levels p	Games
Self-play grid	0, 0.1, ..., 1	550
Rerun, with questions	0.1, 0.5, 0.9	150
No hint	0.1, 0.5, 0.9	150
Neutral wording	0, 0.1, 0.9	150
Neutral wording, goal stated	0, 0.1, 0.9	150
Two rewordings	0.1, 0.9	200
Temperature zero	0.1, 0.9	100
One model, five scripts (4 kinds)	0.1, 0.3, ..., 0.9	1,000
Two models (10 pairs, 1 to 5 seats)	0.1, 0.5, 0.9	1,500
All		3,950

are permutation tests over games, and the supplementary material gives every value behind the statements below.

3.3 Evaluation designs

Table 2 lists the designs; every model plays the same number of games in each.

Self-play and prompt controls. Six copies of one model play the base prompt at eleven risk levels from $p = 0$ to $p = 1$, and again under prompt and decoding controls at fewer risk levels. *No hint* deletes the two equal-split phrases. *Neutral wording* replaces the normative and naming phrases with plain ones, such as “a shared account”. *Goal stated* keeps the neutral wording and adds an opening that says the player is in a paid experiment where “the only thing that matters to you is your own final cash payoff”. Two *rewordings* change the phrasing but keep every rule and the hint, one condition sets the temperature to zero, and a *rerun* repeats the base prompt with value questions.

Value questions. In the rerun, we question the first seat at the start, middle and end of the game about the state in which it has just moved. Each question is a separate call, and its answer never enters the game. Rule questions only ask for numbers printed in the prompt, so we rely on one value question:

Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.

The correct answer is B below p^* , A above it, and equal at p^* . A second value question could be read in two ways, so we report it only in the supplement.

Fixed partners. One model sits beside five scripted partners of one kind. They always pay 0, always pay 2, always pay 4, or copy the group, which means paying 2 in the first round and then the allowed amount nearest to the average of what the other seats paid in the previous round. The model is not told that its partners are

scripted, but their moves are in the history. We compute the *best response* to each kind, the play that earns the most, by enumerating every sequence of the model’s own ten moves. Beside partners who always pay 0, the target cannot be reached; beside partners who always pay 4, it is reached without the model. The best response to both is to pay nothing, and once the history settles the outcome, paying is also dominated. Beside fair-share partners the best response is nothing below p^* and exactly the fair share above it; partners who copy the group never leave 2 and give the same answer.

Mixed tables. To mix defaults at one table, we seat every pair of models together, with one to five seats for the first model, at low, moderate and high risk. Each model’s seats form one block, fixed within a cell and alternating across cells, so every model holds every seat equally often overall.

Selection among models. We finally ask which model prevails when a system keeps or copies the agents that earn more, and how this depends on population size. We treat each model as a strategy in an empirical game [48, 50]. We write $\pi_A(k)$ for the mean expected payoff of an A seat at a table with k seats of A and $6 - k$ of B, from mixed tables for $1 \leq k \leq 5$ and from self-play (grid plus rerun) for $k = 6$. A population of N agents holds i agents of type A and $N - i$ of type B, and each agent’s five tablemates are drawn at random from the other $N - 1$, as in [38]. The fitness of each type is its payoff averaged over the tables it can land in:

$$\begin{aligned} f_A(i) &= \sum_{k=1}^6 \mathcal{H}(k-1; N-1, i-1) \pi_A(k), \\ f_B(i) &= \sum_{k=0}^5 \mathcal{H}(k; N-1, i) \pi_B(6-k), \end{aligned} \quad (2)$$

where $\mathcal{H}(k; N-1, m)$ is the hypergeometric chance that k of five tablemates come from the m other agents of type A. Under Fermi imitation [47], an agent copies a random other agent with probability $1/(1 + e^{-\beta \Delta f})$, where Δf is the other’s fitness minus its own and $\beta = 1$ per payoff unit. We assume rare mutation [17]: each newcomer fixes or disappears before the next arrives. Pairwise fixation probabilities define a Markov chain over the five one-model populations. Its *stationary mass* gives each model’s long-run weight, and EGTtools [13] reproduces the fixation probabilities and stationary distributions of Figure 6. With $N = 6$ the whole population is one table; with $N = 30$, fitness averages over random, overlapping tables. Changing N changes both table compositions and fixation dynamics, not only whom an agent is compared with. We evaluate this finite-population process, not a scoring-only intervention.

4 Results

4.1 Stakes: payment does not track the risk

4.1.1 Self-play across risk. Four of the five models pay about the fair share at every positive risk level (Figure 2a). Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B pay almost the same amount below the pivot as above it (a step of at most 0.2 units), whereas the symmetric welfare benchmark jumps from nothing to the fair share. In the original experiment with the same rules, human groups reached the target in half of their games at $p = 0.9$ and in none at $p = 0.1$ [31]. Strong risk aversion could explain a flat line above zero risk, since a catastrophe leaves nothing and a player with relative risk aversion near one already prefers a

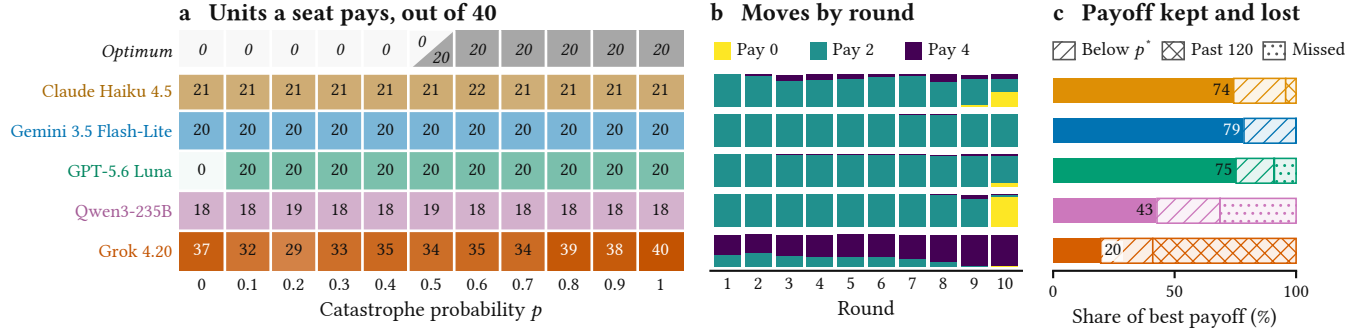


Figure 2: Self-play on the risk grid, 110 games per model. (a) Mean units a seat pays out of 40, ten games per cell; the top row is the optimum of Proposition 1, where both 0 and 20 are optimal at $p = 0.5$. (b) Share of seats paying 0, 2 or 4 in each round, over the games with $p > 0$. (c) Expected payoff kept as a share of $\text{opt}(p)$, and the loss split into paying toward the target below p^* , paying past 120, and missing the target at or above p^* .

certain fair share at low risk. Gemini 3.5 Flash-Lite is the clearest case. Almost every seat plays the exact split at every risk level (Table 3), so a near-fixed rule attains the welfare optimum above the pivot but not below it.

Paying when there is nothing to protect. Four of the five models also pay at $p = 0$, about as much as at any other risk level (Table 3, first column). Paying there only lowers a player’s own money and helps nobody, so neither risk attitude nor concern for others’ cash can explain this payment. GPT-5.6 Luna is the exception. It pays nothing at $p = 0$ and its usual share at every positive risk, so it distinguishes zero risk from positive risk but does not respond to how large the risk is.

Three ways to lose. The models lose payoff in three different ways (Table 3, Figure 2c). Gemini 3.5 Flash-Lite, Claude Haiku 4.5 and GPT-5.6 Luna lose mostly at low risk, where they pay for a target that is not worth reaching. Grok 4.20 overpays: its groups reach the target well before the last round, and its seats keep paying into a pool that is already safe. Qwen3-235B fails at the end. It pays 2 round after round, then almost all of its seats pay nothing in the last round (Figure 2b). Its groups therefore almost never reach the target, although every one of them would have succeeded had each seat simply repeated its previous move. Success rates hide all of this. Gemini 3.5 Flash-Lite and Grok 4.20 both reach the target in every game, yet Grok 4.20 keeps far less of its money. For $0.5 < p < 1$, Qwen3-235B and Grok 4.20 groups earn less in most games than in the worst equilibrium of Proposition 1, in which nobody pays.

Three last-round rules. The last move reveals three rules. Qwen3-235B’s skip follows the clock, not the pool. Across all base-prompt games, it skips the last payment as often when its payment is needed as when the others’ fair shares already suffice (93% of seats in both cases), and it almost never stops earlier once the target is met. Human contributions similarly decline over the rounds of repeated public goods games [3]. GPT-5.6 Luna skips in 2% of cases when the target needs its payment but in 53% when the others’ fair shares would suffice. When six copies each count on another seat to pay, tables ahead of the fair pace of 12 per round often miss the target, while tables exactly on pace reach it. Claude Haiku 4.5 stops paying

Table 3: Self-play over the risk grid. Pays at $p=0$: mean seat total where paying is dominated. Exact split: seats paying 2 in all rounds. Target: games reaching it. Payoff: expected payoff per seat; the optimum averages 25.5. Lost: share of $\text{opt}(p)$ lost. Main loss: paying below p^* , paying past 120, or missing the target at or above p^* (in the last round).

Model	Pays at $p=0$	Exact split (%)	Target (%)	Payoff	Lost Main loss
Claude Haiku 4.5	20.9	17	98	18.9	26 below p^*
Gemini 3.5 Flash-Lite	20.0	99.7	100	20.0	21 below p^*
GPT-5.6 Luna	0.0	72	70	19.2	25 below p^*
Qwen3-235B	18.0	4	2	10.9	57 missed
Grok 4.20	37.4	2	100	5.0	80 past 120

in the last round mostly when the target has already been reached, which shows that the pool can be read from the same prompt.

4.1.2 Prompt and objective controls.

The default survives rewording, but its failures do not. Deleting the two equal-split phrases lowers no model’s payment (Table 4), so the hint was not the source of the payment; it capped some models’ payments and helped others pay exactly 2 per round. Gemini 3.5 Flash-Lite still plays the exact split, which it can rebuild from the target, while GPT-5.6 Luna keeps its average payment but plays the exact split less often. Qwen3-235B and Grok 4.20 pay more without the hint; Qwen3-235B now reaches the target, but it earns less because it overpays. The two rewordings and a temperature of zero leave the payments of the four models other than Grok 4.20 almost unchanged. The failures are less stable than the averages. Under the second rewording Grok 4.20 pays about the fair share, and how often Qwen3-235B drops out in the last round swings widely across prompts. Grok 4.20’s rise with risk on the grid recurs at temperature zero but not in the rerun or the second rewording, so we do not read it as a stable response.

Stating the goal, not removing the norms, cuts payment at zero risk. The normative words are not the main source of the default.

Table 4: Mean seat total under prompt and decoding changes. Top: pooled over $p = 0.1$ and 0.9 (20 games each). Rerun repeats the base prompt with in-game questions; Para. 1, 2 are the rewordings; Wording is the neutral wording and Goal adds the stated goal. Bold: a shift from Base of at least 2 units with permutation $P < 0.01$; every bold shift also has $P < 0.001$, below the Bonferroni threshold for the 35 comparisons. Bottom: single risk levels (10 games each).

Condition	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Base	20.9	20.0	19.9	18.4	35.1
Rerun	21.3	20.0	19.9	18.5	35.6
No hint	21.7	20.0	19.9	34.7	40.0
Para. 1	21.3	20.0	19.9	19.4	31.8
Para. 2	20.9	20.0	19.8	18.0	20.6
Temp. 0	21.2	20.0	20.0	18.3	35.1
Wording	21.0	20.0	19.4	18.7	39.7
Goal	20.0	20.1	13.9	8.2	37.7
Base $p=0$	20.9	20.0	0.0	18.0	37.4
Wording $p=0$	18.0	19.9	0.4	18.1	39.8
Goal $p=0$	0.3	1.4	0.2	4.8	36.7
Goal $p=0.1$	19.8	20.2	7.7	6.4	36.7
Goal $p=0.9$	20.2	20.0	20.1	10.1	38.7

With neutral wording, most models still pay at $p = 0$. At low and high risk the four models other than Grok 4.20 pay on average what they paid before, while Grok 4.20 pays even more and GPT-5.6 Luna plays the exact split less often (Table 4). Adding the stated goal is what changes play at zero risk: Claude Haiku 4.5 and Gemini 3.5 Flash-Lite then pay almost nothing there, Qwen3-235B pays far less than before, and Grok 4.20 barely changes. The goal still does not make most models weigh the risk. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite now separate zero from positive risk but already pay the fair share at low risk, a pattern that strong risk aversion could also produce. Only GPT-5.6 Luna moves from paying little at low risk to the fair share at high risk. Qwen3-235B pays somewhat more at high than at low risk but less than before at every risk level, and never reaches the target; Grok 4.20 still overpays. For most models, payment at zero risk therefore depends on leaving the goal unstated, while the flat response above zero risk does not. The partner and selection results below use the base prompt.

4.1.3 Correct answers, unchanged play. Every rule question is answered correctly. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite answer the value question correctly almost every time, and GPT-5.6 Luna most of the time. All three change their answer with the risk (Figure 3a): at low risk they almost always say that keeping pays more, and at high risk they mostly say the opposite. Their payments barely change (Figure 3b). Qwen3-235B always says that keeping pays more, and Grok 4.20 gives that answer in the first round, whatever the risk. The question supplies two hypothetical outcomes rather than the partners’ responses to a deviation, and it makes the comparison explicit, unlike the move request. Correct answers therefore do not establish knowledge of a best response, and the result is a gap between answers and play, not evidence about what an agent computes. A strongly risk-averse agent could give these

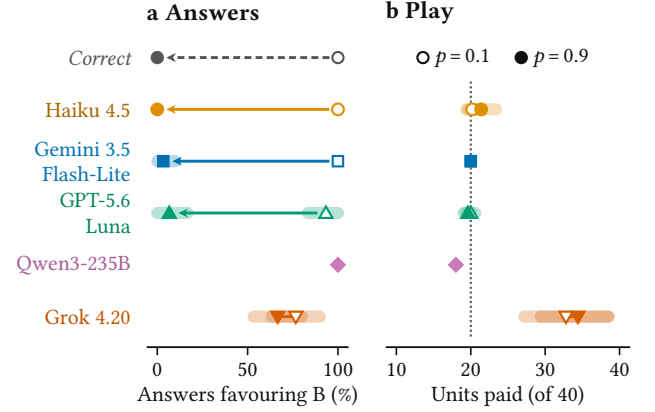


Figure 3: Answers and play at risk 0.1 (open) and 0.9 (filled), ten games per model and risk; arrows run from low to high risk, and shaded bars are 95% game-bootstrap intervals. (a) Share of answers favouring B; dashed: a fully correct switch. (b) Units the questioned seat pays out of 40; dotted: fair share. This compares answers and actions, not best responses; the supplement gives the shifts with their intervals and all three risks.

answers and still pay; payment at $p = 0$ and after settlement is the sharper test.

4.2 Partners: simple responses and fragile tables

4.2.1 Fixed partners.

No model keeps all its money where that is the best response. In no game beside partners who always pay 0 or always pay 4 does the model pay nothing. Only 186 of the 1,000 scripted games are exactly optimal, and every one of them is a game in which the optimum is the fair share. The models land on the optimum only where the optimum is the default. An agent that does not know its partners are scripted may reasonably pay early to learn who they are, so the cleaner evidence is payment after the outcome is settled (Table 5).

Partners matter; risk does not. We split the variation in what a model pays beside the three fixed-amount partners into partner, risk and interaction parts. For Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B, the kind of partner explains most of the variation (69% to 92%) and the risk with its interaction at most 5%. For an exact best responder in the same games, the risk would explain half of it. Beside fair-share partners, where the best response jumps from nothing to the fair share at p^* , no model’s total moves by more than a single payment of 2 between low and high risk. Grok 4.20’s total follows neither: it is set mostly by its opening move, made before any history exists, which is the largest payment in about half of its games and which it mostly repeats.

Responses to partners take simple forms. The models respond to their partners, but not as a best responder would (Table 5; the supplementary material shows every round). Gemini 3.5 Flash-Lite pays its usual rate; both it and Claude Haiku 4.5 stop once the history shows that the outcome is settled. Beside partners who

Table 5: Units a model pays per game before and after the outcome is settled, beside five partners who always pay 0 or always pay 4 (50 games per entry). Given these partners, every unit paid is wasted; units paid after settlement are wasted even for an agent unaware that its partners are scripted.

Model	Always 0		Always 4	
	Before	After	Before	After
Claude Haiku 4.5	7.7	0.1	14.2	1.1
Gemini 3.5 Flash-Lite	13.6	0.2	10.9	0.0
GPT-5.6 Luna	8.5	0.0	4.5	0.0
Qwen3-235B	17.8	12.1	12.3	8.1
Grok 4.20	19.6	11.1	16.7	16.2

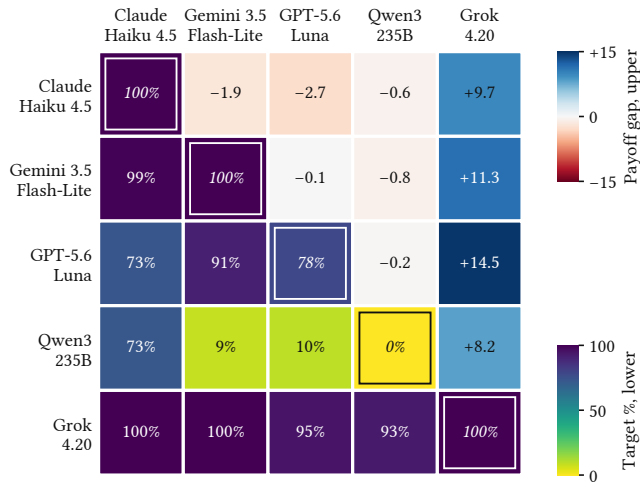


Figure 4: Two models at one table, 150 games per pair, pooled over seat splits and risk. Above the diagonal: expected payoff of the row model’s seats minus the column model’s, in the same games. On and below it: share reaching the target; boxed: self-play (60 games, grid plus rerun at the same risks).

always pay 4, GPT-5.6 Luna pays little even before that point, close to a best response while it learns who its partners are. Qwen3-235B pays more, not less, beside partners who pay nothing, and both Qwen3-235B and Grok 4.20 keep paying large amounts after the outcome is settled, which no belief about the partners can justify. Beside five fair-share partners, Qwen3-235B pays 2 for nine rounds and then nothing, so the table misses the target in almost every game by that single payment; Gemini 3.5 Flash-Lite in the same seat reaches it every time.

4.2.2 Mixed tables. At a shared table every model earns more than Grok 4.20, while the payoff gaps among the other four models are small (Figure 4, above the diagonal). This follows from Proposition 2: Grok 4.20 pays the most, and at one table the larger payer earns less. Group success depends on the pair (Figure 4, below the diagonal). Tables that include Grok 4.20 reach the target in the great majority of games, while tables that mix Qwen3-235B with Gemini 3.5 Flash-Lite or GPT-5.6 Luna rarely do.

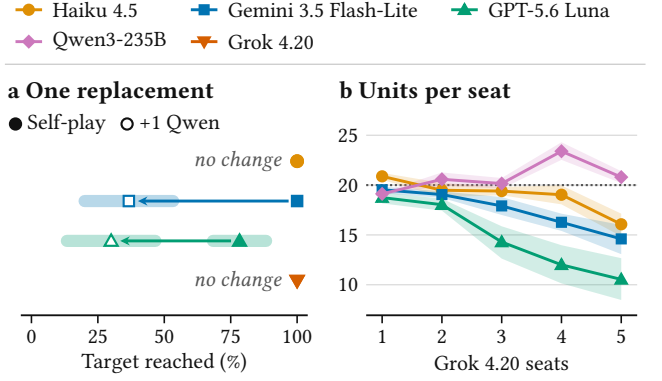


Figure 5: Mixed groups, pooled over $p = 0.1, 0.5, 0.9$. (a) Self-play (filled; 60 games, grid plus rerun) versus one Qwen3-235B replacement (open; 30); arrows run from the first to the second. (b) Mean contribution per focal seat beside one to five Grok 4.20 seats (30 games each); dotted: fair share. Bars and bands: 95% game-bootstrap intervals, not stratified by risk. Lines join measured means.

An exact fair share has no slack. One partner that stops paying in the last round can break a table of exact fair-share players (Figure 5a). When the pool is one round of fair shares short before the last round, Qwen3-235B pays nothing in 92% of cases and Gemini 3.5 Flash-Lite in 0.1%. Five Gemini 3.5 Flash-Lite seats with one Qwen3-235B seat succeed in only 11 of 30 games, although Gemini 3.5 Flash-Lite alone never misses in our sample; tables with GPT-5.6 Luna also succeed less often. Tables of Claude Haiku 4.5 or Grok 4.20, which pay more, absorb the same partner. Replacing *all* Qwen3-235B last-round zeros by 2, holding other recorded actions fixed, would rescue 92% of the 323 failed mixed tables containing it. This repair is arithmetic, not a tested intervention. Conversely, one Grok 4.20 seat lets five Qwen3-235B seats succeed in most games and pays for the rescue. These comparisons depend on the base prompt and the fixed seat blocks.

Agents free-ride on generous partners. Most agents pay less as the number of Grok 4.20 seats rises from one to five (Figure 5b). Beside five, GPT-5.6 Luna pays 10.5 units, and Gemini 3.5 Flash-Lite and Claude Haiku 4.5 also fall below the fair share; these two stop once the pool reaches the target, while GPT-5.6 Luna pays less from the start. Qwen3-235B does not show this decline. Within a cell, Grok 4.20’s higher opening payments are associated with lower total payments by Gemini 3.5 Flash-Lite and Claude Haiku 4.5, mainly through earlier target completion. GPT-5.6 Luna also pays less while the pool is short, consistent with the compensation predicted beside artificial agents [44]. Openings are not randomised and partners can react to a deviation, so this association identifies neither a causal response nor an exact best response.

4.3 Selection: population size changes which model is favoured

In a population of one table, selection rewards the smallest payer. Within one game, Proposition 2 makes the ranking mechanical: the

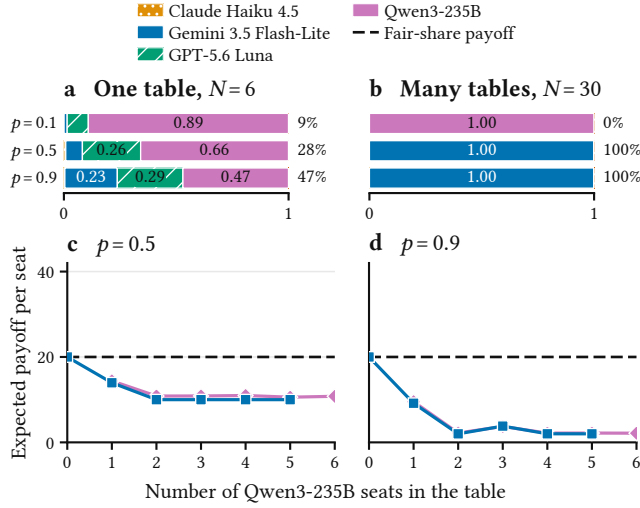


Figure 6: Rare-mutation Fermi selection ($\beta = 1$). (a, b) Stationary mass in a population of one table ($N = 6$) or a larger randomly mixed population ($N = 30$); right of each bar, the self-play target rate weighted by mass. (c, d) Expected payoff of a Qwen3-235B or Gemini 3.5 Flash-Lite seat by the number of Qwen3-235B seats at its table, at $p = 0.5$ and $p = 0.9$; dashed: fair-share payoff. Invasion diagrams are in the supplement.

seat that pays less earns more. What the data add is who pays least in company, which is Qwen3-235B, and what this selection costs the group. Selection gives Qwen3-235B the largest weight at every risk level (Figure 6a). Reaching the target is one of two optima at $p = 0.5$ and the only one at $p = 0.9$. At both risks, the selected population reaches it far less often than a model picked at random (47% against 76% at $p = 0.9$), and stronger selection pushes group success close to zero. A worked example shows the mechanism. Beside five exact fair-share seats at $p = 0.9$, a seat that skips its last payment makes the table miss, so its expected payoff falls from 20 to 2.2, yet it still earns more than each tablemate, who expects 2. This outcome is the known result that selection on payoff relative to a small group favours whoever does better than its neighbours, even at a cost to the group [39], here shown with deployed language models.

A larger population favours the fair-share player. In the larger population, selection favours Gemini 3.5 Flash-Lite at moderate and high risk (Figure 6b). Its self-play attains the welfare optimum there, but neither optimality nor Nash equilibrium implies that imitation selects it. The ranking comes from the estimated payoffs at every composition along the fixation path, where a lower payer can earn more than its tablemates even while lowering its own payoff (Figure 6c, d). At low risk both population sizes favour Qwen3-235B, still far below the optimum. The switch to Gemini 3.5 Flash-Lite occurs already at seven or eight agents, consistent with finite-population theory [39]. Each winner stays on top in at least 93% of bootstrap samples, and the same winners appear at $\beta = 10$, at N of 60 and 120 and with grid-only self-play payoffs. Under weak selection ($\beta = 0.1$), the largest one-table mass is only 0.34 and GPT-5.6 Luna narrowly leads at moderate and high risk. Which

model wins remains conditional on the base-prompt payoffs, whose signatures change with wording.

Post-hoc checks show that the winners depend on the candidate set: without Qwen3-235B, GPT-5.6 Luna wins the one-table population at all tested risks, and without Gemini 3.5 Flash-Lite, it wins the larger population at moderate and high risk. The supplement reports every omission and the mass-weighted welfare.

5 Discussion and conclusion

Possible causes. Our data record choices, not reasoning, so we offer hypotheses. The equal split may be a focal point that models reconstruct without the hint [10]. A question may call up computation while a move request calls up a default [12, 40]. Post-training may favour doing one’s share when nothing says otherwise, consistent with models being more cooperative than people [30] and cooperating more as the stakes of a repeated prisoner’s dilemma grow, against the evolutionary prediction [25]. Stronger reasoning is not an obvious fix, since reasoning models can free-ride more in public goods games [18].

Implications for multiagent evaluation. State the agent’s goal, since stating it cut zero-risk payment for most models. Report $\text{opt}(p)$, a dominated-action condition and payment after settlement, since success can coexist with wasted money, much as cooperation and welfare can diverge in evolutionary models [21]. Test mixed populations, where an exact fair share can fail beside one late dropout. Audit how population size and the imitation process affect selection, since a payoff advantage over tablemates need not improve group welfare.

Threats, limitations and ethics. The goal sentence is an instruction tested in one phrasing and in self-play only, and without a group-payoff goal we cannot separate objective clarity from the direction of the instruction. The running pool is not printed, so arithmetic errors may explain some payment after settlement, and the value questions omit $p = 0$. We test five models, one per provider, in English and one game; claims concern this panel, not language models in general. Ten games per cell limit power, and a zero-width bootstrap interval in a constant sample does not prove deterministic behaviour. Rewordings cover a small prompt space, welfare losses assume risk neutrality, the scripted, mixed and selection designs use the base prompt, fixed seat blocks confound composition with position, and selection assumes rare entry. We will release prompts, configurations, seeds and code; they reproduce the design, not the model transcripts. The study involves no human subjects or personal data, and we report failure modes so designers can test for them before trusting agents with shared resources.

Conclusion. Cooperative-looking self-play can come from a default that ignores the stakes. For the five models we test, this default fails where paying is dominated, beside a single late dropout, and under selection in a population of one table. Several of these models rank the two choices correctly when asked, yet keep paying the same. Varying the stakes, fixing the partners and mixing the population expose this default; self-play success alone does not. Whether a group-payoff goal or a printed pool moves it remains open.

References

- [1] Maria Abou Chakra and Arne Traulsen. 2012. Evolutionary dynamics of strategic behavior in a collective-risk dilemma. *PLoS Computational Biology* 8, 8 (2012), e1002652. doi:10.1371/journal.pcbi.1002652
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] James Andreoni. 1988. Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics* 37, 3 (1988), 291–304. doi:10.1016/0047-2727(88)90043-6
- [4] Alessio Buscemi, Daniele Proverbio, Paolo Bova, Nataliya Balabanova, Adeela Bashir, Theodor Cîmpeanu, Henrique Correia da Fonseca, Manh Hong Duong, Elias Fernández Domingos, Antonio M. Fernandes, Marcus Krellner, Ndidi Bianca Ogbo, Simon T. Powers, Fernando P. Santos, Zia Ush Shamszaman, Zhao Song, Alessandro Di Stefano, and The Anh Han. 2025. Do LLMs trust AI regulation? Emerging behaviour of game-theoretic LLM agents. arXiv:2504.08640 doi:10.48550/arXiv.2504.08640
- [5] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. 2025. FAIRGAME: A Framework for AI Agents Bias Recognition Using Game Theory. In *ECAI 2025 – 28th European Conference on Artificial Intelligence (Frontiers in Artificial Intelligence and Applications, Vol. 413)*. IOS Press, Bologna, Italy, 4097–4104. doi:10.3233/FAIA251300
- [6] Luis Celiktemel, Edward Eichhorn, Levin Brinkmann, Robin Schimmelpfennig, Aron Vallinder, Yaomin Jiang, Edward Hughes, and Iyad Rahwan. 2026. Group Selection Promotes Prosocial Prompts in Populations of LLM Agents. arXiv:2606.23343 doi:10.48550/arXiv.2606.23343
- [7] Alan Chan, Carson Ezell, Max Kaufmann, Kevin Wei, Lewis Hammond, Herbie Bradley, Emma Bluemke, Nitarshan Rajkumar, David Krueger, Noam Kolt, Lennart Heim, and Markus Anderljung. 2024. Visibility into AI Agents. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, Rio de Janeiro, Brazil, 958–973. doi:10.1145/3630106.3658948
- [8] Yiting Chen, Tracy Xiao Liu, You Shan, and Songfa Zhong. 2023. The emergence of economic rationality of GPT. *Proceedings of the National Academy of Sciences* 120, 51 (2023), e2316205120. doi:10.1073/pnas.2316205120
- [9] Allan Dafoe, Edward Hughes, Yoram Bachrach, Tatum Collins, Kevin R. McKee, Joel Z. Leibo, Kate Larson, and Thore Graepel. 2020. Open Problems in Cooperative AI. arXiv:2012.08630 doi:10.48550/arXiv.2012.08630
- [10] Éléonore Dodivers and Ismaël Rafai. 2025. Uncovering the fairness of AI: exploring focal point, inequality aversion, and altruism in ChatGPT’s dictator game decisions. *Economics Bulletin* 45, 4 (2025), 1818–1825.
- [11] M. H. Duong, C. M. Durbac, and T. A. Han. 2026. Cost of Institutional Incentives for Promoting Cooperation in 2 × 2 Games and Collective Risk Games. *Dynamic Games and Applications*. Online first, 7 February 2026. doi:10.1007/s13235-025-00696-6
- [12] Caoyun Fan, Jindou Chen, Yaohui Jin, and Hao He. 2024. Can Large Language Models Serve as Rational Players in Game Theory? A Systematic Analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. AAAI Press, Vancouver, Canada, 17960–17967. doi:10.1609/aaai.v38i16.29751
- [13] Elias Fernández Domingos, Francisco C. Santos, and Tom Lenaerts. 2023. EGT-tools: Evolutionary game dynamics in Python. *iScience* 26, 4 (2023), 106419. doi:10.1016/j.isci.2023.106419
- [14] Elias Fernández Domingos, Inés Terrucha, Rémi Suchon, Jelena Grujić, Juan C. Burguillo, Francisco C. Santos, and Tom Lenaerts. 2022. Delegation to artificial agents fosters prosocial behaviors in the collective risk dilemma. *Scientific Reports* 12 (2022), 8492. doi:10.1038/s41598-022-11518-9
- [15] Apostolos Filippas, John J. Horton, and Benjamin S. Manning. 2024. Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?. In *Proceedings of the 25th ACM Conference on Economics and Computation (EC ’24)*. ACM, New Haven, CT, USA, 614–615. doi:10.1145/3670865.3673513
- [16] Nicolò Fontana, Francesco Pierri, and Luca Maria Aiello. 2025. Nicer than Humans: How Do Large Language Models Behave in the Prisoner’s Dilemma? *Proceedings of the International AAAI Conference on Web and Social Media* 19 (2025), 522–535. doi:10.1609/icwsm.v19i1.35829
- [17] Drew Fudenberg and Loren A. Imhof. 2006. Imitation processes with small mutations. *Journal of Economic Theory* 131, 1 (2006), 251–262. doi:10.1016/j.jet.2005.04.006
- [18] David Guzman Piedrahita, Yongjin Yang, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, and Zhijiang Jin. 2025. Corrupted by Reasoning: Reasoning Language Models Become Free-Riders in Public Goods Games. In *Conference on Language Modeling (COLM)*. OpenReview.net, Montreal, Canada, 37 pages. https://openreview.net/forum?id=kh6LOHGJel
- [19] Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, Tomáš Gavenčák, The Anh Han, Edward Hughes, Vojtěch Kovařík, Jan Kulveit, Joel Z. Leibo, Caspar Oesterheld, Christian Schroeder de Witt, Nisarg Shah, Michael Wellman, Paolo Bova, Theodor Cîmpeanu, Carson Ezell, Quentin Feuillade-Montixi, Matija Franklin, Esben Kran, Igor Krawczuk, Max Lamparth, Niklas Lauffer, Alexander Meinke, Sumeet Motwani, Anka Reuel, Vincent Conitzer, Michael Dennis, Iason Gabriel, Adam Gleave, Gillian Hadfield, Nika Haghtalab, Atoosa Kasirzadeh, Sébastien Krier, Kate Larson, Joel Lehman, David C. Parkes, Georgios Piliouras, and Iyad Rahwan. 2025. Multi-Agent Risks from Advanced AI. Cooperative AI Foundation, Technical Report #1. arXiv:2502.14143 doi:10.48550/arXiv.2502.14143
- [20] The Anh Han. 2022. Emergent behaviours in multi-agent systems with Evolutionary Game Theory. *AI Communications* 35, 4 (2022), 327–337. doi:10.3233/AIC-220104
- [21] The Anh Han, Manh Hong Duong, and Matjaž Perc. 2024. Evolutionary mechanisms that promote cooperation may not promote social welfare. *Journal of the Royal Society Interface* 21, 220 (2024), 20240547. doi:10.1098/rsif.2024.0547
- [22] Jen-tse Huang, Eric John Li, Man Ho Lam, Tian Liang, Wenxuan Wang, Youliang Yuan, Wenxiang Jiao, Xing Wang, Zhaopeng Tu, and Michael R. Lyu. 2025. Competing Large Language Models in Multi-Agent Gaming Environments. In *International Conference on Learning Representations (ICLR)*. OpenReview.net, Singapore, 65639–65681.
- [23] Edward Hughes, Joel Z. Leibo, Matthew Phillips, Karl Tuyls, Edgar Dueñez-Guzman, Antonio García Castañeda, Iain Dunning, Tina Zhu, Kevin McKee, Raphael Koster, Heather Roff, and Thore Graepel. 2018. Inequity aversion improves cooperation in intertemporal social dilemmas. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates, Inc., Montreal, Canada, 11 pages.
- [24] Trung-Kiet Huynh et al. 2025. Understanding LLM Agent Behaviours via Game Theory: Strategy Recognition, Biases and Multi-Agent Dynamics. arXiv:2512.07462 doi:10.48550/arXiv.2512.07462
- [25] Trung-Kiet Huynh et al. 2026. Payoff scaling shapes cooperation in LLM agents across languages. arXiv:2601.19082 doi:10.48550/arXiv.2601.19082
- [26] Shivani Kumar, Adarsh Bharathwaj, and David Jurgens. 2026. Cooperative Profiles Predict Multi-Agent LLM Team Performance in AI for Science Workflows. arXiv:2604.20658 doi:10.48550/arXiv.2604.20658
- [27] Joel Z. Leibo, Edgar A. Dueñez-Guzman, Alexander Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charlie Beattie, Igor Mordatch, and Thore Graepel. 2021. Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, Virtual Event, 6187–6199. https://proceedings.mlr.press/v139/leibo21a.html
- [28] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. 2017. Multi-agent Reinforcement Learning in Sequential Social Dilemmas. In *Proceedings of the 16th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2017)*. IFAAMAS, São Paulo, Brazil, 464–473. https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf
- [29] Nunzio Lorè and Babak Heydari. 2024. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports* 14 (2024), 18490. doi:10.1038/s41598-024-69032-z
- [30] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O. Jackson. 2024. A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences* 121, 9 (2024), e2313925121. doi:10.1073/pnas.2313925121
- [31] Manfred Milinski, Ralf D. Sommerfeld, Hans-Jürgen Krambeck, Floyd A. Reed, and Jochem Marotzke. 2008. The collective-risk social dilemma and the prevention of simulated dangerous climate change. *Proceedings of the National Academy of Sciences* 105, 7 (2008), 2291–2294. doi:10.1073/pnas.0709546105
- [32] Shayegan Omidshafiei, Christos Papadimitriou, Georgios Piliouras, Karl Tuyls, Mark Rowland, Jean-Baptiste Lespiau, Wojciech M. Czarnecki, Marc Lanctot, Julien Perolat, and Remi Munos. 2019. α -Rank: Multi-Agent Evaluation by Evolution. *Scientific Reports* 9, 1 (2019), 9937. doi:10.1038/s41598-019-45619-9
- [33] Saptarshi Pal, Abhishek Mallela, Lenz Pracher, Chiyu Wei, Feng Fu, Santiago Schnell, and Martin A. Nowak. 2026. Large language models instantiate evolutionarily robust strategies of cooperation. *PNAS Nexus* 5, 6 (2026), pgag210. doi:10.1093/pnasnexus/pgag210
- [34] Thomas R. Palfrey and Howard Rosenthal. 1984. Participation and the provision of discrete public goods: a strategic analysis. *Journal of Public Economics* 24, 2 (1984), 171–193. doi:10.1016/0047-2727(84)90023-9
- [35] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 doi:10.48550/arXiv.2507.02618
- [36] Phu Hoa Pham et al. 2026. Humans Are More Diverse: Frontier LLMs Show Extreme Policies in Idealised AI Development Races. arXiv:2608.01193 doi:10.48550/arXiv.2608.01193
- [37] Giorgio Piatti, Zhijiang Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Vancouver, Canada, 111715–111759. doi:10.52202/079017-3548

- [38] Francisco C. Santos and Jorge M. Pacheco. 2011. Risk of collective failure provides an escape from the tragedy of the commons. *Proceedings of the National Academy of Sciences* 108, 26 (2011), 10421–10425. doi:10.1073/pnas.1015648108
- [39] Mark E. Schaffer. 1988. Evolutionarily stable strategies for a finite population and a variable contest size. *Journal of Theoretical Biology* 132, 4 (1988), 469–478. doi:10.1016/S0022-5193(88)80085-7
- [40] Thomas Schmied, Jörg Bornschein, Jordi Grau-Moya, Markus Wulfmeier, and Razvan Pascanu. 2026. LLMs are Greedy Agents: Effects of RL Fine-tuning on Decision-Making Abilities. In *International Conference on Learning Representations (ICLR)*. OpenReview.net, Rio de Janeiro, Brazil, 127538–127571. <https://openreview.net/forum?id=weUP6H5Ko9>
- [41] Oliver Slumbers, Joel Z. Leibo, and Marco A. Janssen. 2025. Using Large Language Models to Simulate Human Behavioural Experiments: Port of Mars. arXiv:2506.05555 doi:10.48550/arXiv.2506.05555
- [42] Peter Stone, Gal A. Kaminka, Sarit Kraus, and Jeffrey S. Rosenschein. 2010. Ad Hoc Autonomous Agent Teams: Collaboration without Pre-Coordination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 24. AAAI Press, Atlanta, Georgia, USA, 1504–1509. doi:10.1609/aaai.v24i1.7529
- [43] Alessandro Tavoni, Astrid Dannenberg, Giorgos Kallis, and Andreas Löschel. 2011. Inequality, communication, and the avoidance of disastrous climate change in a public goods game. *Proceedings of the National Academy of Sciences* 108, 29 (2011), 11825–11829. doi:10.1073/pnas.1102493108
- [44] Inês Terrucha, Elias Fernández Domingos, Francisco C. Santos, Pieter Simoens, and Tom Lenaerts. 2024. The art of compensation: How hybrid teams solve collective-risk dilemmas. *PLOS ONE* 19, 2 (2024), e0297213. doi:10.1371/journal.pone.0297213
- [45] Emanuel Tewolde, Xiao Zhang, David Guzman Piedrahita, Vincent Conitzer, and Zhijiang Jin. 2026. CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. PMLR, Seoul, South Korea, 65 pages. <https://arxiv.org/abs/2604.15267>
- [46] Arne Traulsen and Martin A. Nowak. 2006. Evolution of cooperation by multilevel selection. *Proceedings of the National Academy of Sciences* 103, 29 (2006), 10952–10955. doi:10.1073/pnas.0602530103
- [47] Arne Traulsen, Martin A. Nowak, and Jorge M. Pacheco. 2006. Stochastic dynamics of invasion and fixation. *Physical Review E* 74, 1 (2006), 011909. doi:10.1103/PhysRevE.74.011909
- [48] Karl Tuyls, Julien Perolat, Marc Lanctot, Joel Z. Leibo, and Thore Graepel. 2018. A Generalised Method for Empirical Game Theoretic Analysis. In *Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2018)*. IFAAMAS, Stockholm, Sweden, 77–85. <https://ifaamas.org/Proceedings/aamas2018/pdfs/p77.pdf>
- [49] Aron Vallinder and Edward Hughes. 2025. Cultural Evolution of Cooperation among LLM Agents: Extended Abstract. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*. IFAAMAS, Detroit, MI, USA, 2771–2773. <https://www.ifaamas.org/Proceedings/aamas2025/pdfs/p2771.pdf>
- [50] Michael P. Wellman. 2006. Methods for Empirical Game-Theoretic Analysis (Extended Abstract). In *Proceedings of the Twenty-First National Conference on Artificial Intelligence (AAAI-06)*. AAAI Press, Boston, Massachusetts, USA, 1552–1555.
- [51] Richard Willis, Jianing Zhao, Yali Du, and Joel Z. Leibo. 2026. Evaluating Collective Behaviour of Hundreds of LLM Agents. Strategic Engineering Workshop on LLMs and Game Theory (SE@AAMAS 2026). arXiv:2602.16662 doi:10.48550/arXiv.2602.16662